Software Requirements Specification (SRS)

Hospital Booking System for the Outpatient Department (OPD) | Team Aura | Version 1.0 - revised draft

Document Control

Field	Value
Team name	Team Aura
Project title	The Hospital Booking System
Version	1.0 -revised draft
SDLC	Incremental / Iterative (or Agile)
GitHub repository	https://github.com/Min-Hein-Ko/The-hospital-booking-system.git

1. Introduction

1.1 Purpose and Scope

The purpose of the Hospital Booking System for the Outpatient Department (OPD) is to allow patients to view doctor information, specializations and available schedules, book appointments online, receive token numbers, and check the current token queue. The system also supports administrators in managing doctors, schedules, appointments, and patient queues.

1.2 Users and Stakeholders

Role	Type	Needs / Responsibilities
User (Patient)	Primary system user	Register/login, view doctor information and availability, book appointments, receive a token number, check the current queue, and cancel or reschedule appointments.
Admin	System role	Manage doctor information, schedules and appointment availability, ban or restore patient account access.
Doctor	Secondary stakeholder	Needs accurate schedule and appointment information. The current system does not define a doctorfacing role.

2. Overall Description

2.1 Product Context

The product is a web-based system for outpatient appointment booking.The defined system roles are User (Patient) and Admin.

Patient flow: register/login → view doctor information and availability → select a doctor, date, and time → book an appointment → receive a token number → have an appointment. Admins manage doctor information, schedules, appointments, and tokens; they can call the next patient and update token status.

2.2 In-Scope Functions

FR-1: Patients view doctor information and availability.

FR-2: Patients book appointments, provide optional visit reasons, and receive tokens.

FR-3: Patients view their own token numbers and appointment statuses.

FR-4: Patients cancel, reschedule, or arrange follow-up appointments.

FR-5: Admins manage doctor information and schedules.

FR-6: Admins monitor dashboards, appointments, and token records.

FR-7: Admins ban or restore patient accounts.

FR-8: Patients register, log in, and manage their profiles.

FR-9: Patients and Admins receive and manage notifications.

2.3 Out-of-Scope Functions

- Online payment or billing.
- Online medical consultation or video calls.
- Medical diagnosis or treatment recommendations.
- Prescription management and electronic medical records management.
- Emergency or ambulance services.
- Insurance claim processing.
- Online pharmacy or medicine ordering.
- Integration with external hospital systems.
- AI-based medical diagnosis.

2.4 Assumptions and Constraints

- Patients can access the system through the internet using an electronic device.
- The Admin keeps doctor information, specializations, department assignments, and available schedules up to date.
- Users provide personal information when registering and making appointments (unnecessary personal data are not needed)
- Users must have an account and be logged in before booking, changing, or cancelling an appointment.
- Only Admins can add, edit, freeze, or remove doctor profiles.
- Only Admins can call patients and update token status.
- Every confirmed appointment receives a system-generated token number.
- Token statuses are Waiting, Called, In Operation, Completed, and Cancelled.

3. Functional Requirements

3.1 FR-1 — View Doctor Information and Availability

The system shall allow all the users (patients) to view all the active doctors, their specializations and degrees, departments, qualifications from various majors, profiles, and available appointment schedules within the website.

Acceptance Criteria:

1. The system displays all active doctors and their specialty.
2. Patients can view available appointment dates and time for the booking.

3. Unavailable doctors who have already booked and unavailable time slots cannot be selected.

3.2 FR-2 — Book an Appointment and Receive a Token

The system shall allow patients to book available appointments and receive an automatically generated token number immediately after a successful booking.33

Acceptance Criteria:

1. The system allows active patients to select an available doctor, date, and time.
2. After successful booking, the system automatically generates a unique token with an initial status of waiting and prevents duplicate bookings from other users.

3.3 FR-3 — View Personal Token and Appointment Status

The system shall allow all patients to view their personal token numbers, appointment details with accurate date and time, and latest appointment statuses.

cceptance Criteria:

1. The system displays the patient's token number, doctor, appointment date and time.
2. The system updates the displayed status when the backend has a valid status change.
3. Patients can only view their own personal token and appointment status.

3.4 FR-4 — Cancel or Reschedule an Appointment and Arrange a Follow-up

The system shall allow patients who have already booked to cancel or reschedule their eligible appointments and arrange follow-up appointments after completing a visit.

Acceptance Criteria:

1. The system allows patients to cancel the upcoming appointments and updates their status to Cancelled.
2. Patients can reschedule their appointments to another available time, and the system generates a new token immediately.
3. Follow-up bookings create new appointment records without overwriting completed appointments.

3.5 FR-5 — Manage Doctor Information and Schedules

The system shall allow the website Admins to add, update, and deactivate doctor profiles when they have left the hospital and manage all their schedules.

Acceptance Criteria:

1. Admins can edit and update doctor information and their schedules.
2. Updated doctor information and available schedules appear from the patient's point of view.
3. The system prevents permanent deletion of doctors with existing appointment records.

3.6 FR-6 — View Admin Dashboard, Appointments, and Token Records

The system shall allow Admins to monitor all the hospital statistics, appointment information, and token records.

Acceptance Criteria:

1. The dashboard displays appointment statistics, active doctors, and patient personal account information.
2. Admins can view, search, and filter appointments and token records from their side.
3. Appointment statuses and token information are read-only in their dashboard, so Admins cannot manually control the queue.

3.7 FR-7 — Manage Patient Account Access

The system shall allow Admins to view patient accounts, ban accounts, and restore patient access.

Acceptance Criteria:

1. Admins can view patient information and account statuses (Active or Banned).
2. Admins can ban or restore patient accounts, and the system records these actions.
3. Banning an account block booking and rescheduling their appointments.

3.8 FR-8 — Register, Sign In, and Manage Patient Profile

The system shall allow the patients (users) to register, sign in, and manage their personal profile information.

Acceptance Criteria:

1. The system allows new patients to sign up the new account and existing patients to sign in.
2. Patients can view and edit their own personal profile information.
3. Banned patients cannot access appointment booking or rescheduling services until the admin team releases them.

3.9 FR-9 — Receive and Manage In-App Notifications

The system shall allow patients and Admins to receive and manage notifications about appointments, account activities, and system events.

Acceptance Criteria:

1. The system displays notifications and unread counts for each user's account.
2. Users can view notifications and mark each or all notifications as read.
3. The system sends relevant notifications when appointments, account statuses, or appointment statuses change. Notifications must not contain visit-reason information.

4. Non-Functional Requirements

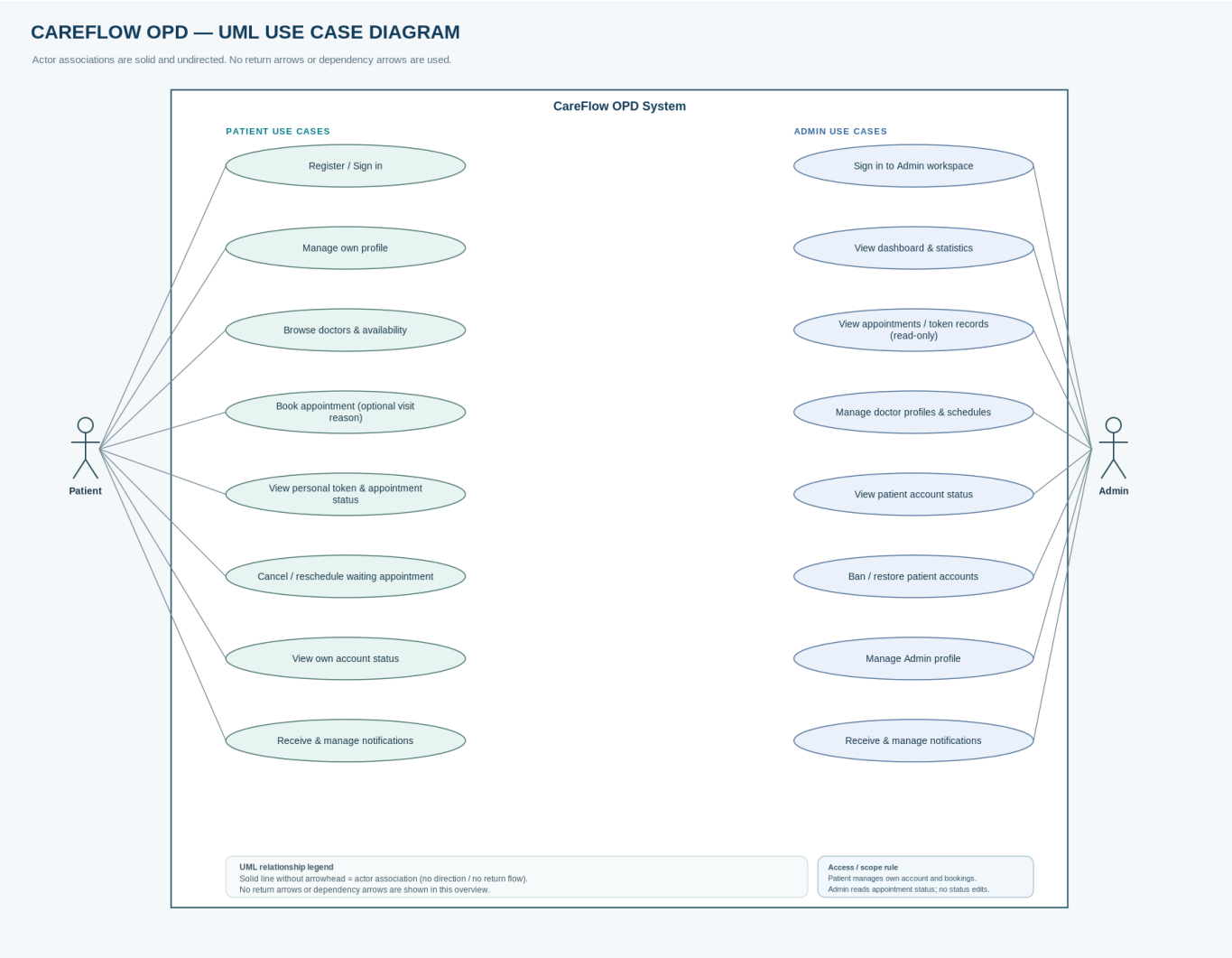
ID / Type	Requirement	Verification
NFR-1 Performance	Doctor and appointment schedule pages shall load within 3 seconds under normal network conditions.	Measure page load time under normal network conditions.
NFR-2 Performance	The system shall create and display a patient's token number within 2 seconds after booking confirmation.	Measure the time between booking confirmation and token display.

NFR-3 Usability	A first-time user shall be able to find a doctor and complete an appointment booking in under 2 minutes without assistance.	Conduct a timed trial with a first-time user.
NFR-4 Reliability	The system shall save a confirmed appointment and its token number so that the records remain available after the user leaves or refreshes the appointment page.	Create an appointment, refresh the page, and verify the appointment and token records.
NFR-5 Security / Privacy	A user shall only be able to view and manage their own personal information and booking history.	Log in as one user and attempt to access another user's information; access must be prevented.

5. Use Cases

Use Case	Actor	Related FR	Brief Success Outcome
UC-1 View the doctor information	User (Patient)	FR-1	The patient can view the doctor details, specializations and their available schedue.
UC-2 Book appointment	User (Patient)	FR-2	Appointment is booked and a unique token is generated after a successful booking.
UC-3 View their appointment status	User (Patient)	FR-3	Patient views their token and latest appointment status.
UC-4 Manage appointment	User (Patient)	FR-4	Patients can successfully cancel, reschedule, or book a follow-up appointment.
UC-5 Manage doctor and schedules	Admin	FR-5	Doctor information and schedules can be updated successfully. The admin updates doctor/schedule information manages appointments.
UC-6 Monitor Dashboard and Records	Admin	FR-6	Admin can view dashboard statistics, appointments, and token records.
UC-7 Manage Patient Accounts	Admin	FR-7	Admin can ban, freeze or restore the patients' accounts.
UC-8 Manage Patient Profile	Patient	FR-8	Patients can sign up new accounts or sign in their accounts. They can also update their profile information.

UC-9 Manage notifications	Patient and Admin	FR-9	All the users and admins can view their notifications and read their status.



6. Token Status Model

The token statuses specified in the project materials are Waiting, Called, In Operation, Completed, and Cancelled. The exact token-number assignment algorithm (for example, the ordering rule) has not been defined in the available materials and requires team confirmation.

7.Traceability Summary (Golden Thread)

Problem / Need	Requirement	design	Feature	Test
Patients need to find doctors and check availability.	FR-1	SC-04 Find a Doctor	Doctor search, specialty filter, active doctor list, and available appointment slots.	**Pass**: Active doctors, specialties, available dates and times were displayed. Unavailable doctors and booked time slots could not be selected.
Patients need to book appointments and receive a token.	FR-2, FR-3	SC-04 Find a Doctor → SC-06 Token Queue	Book an appointment, provide an optional visit reason, generate a token, and display booking confirmation.	**Partial Pass**: Patients could select available booking options and received a token after successful booking. Token uniqueness and duplicate-booking prevention were not fully verified.
Patients need to know their queue progress.	FR-3	SC-06 Token Queue	Display the patient's own token number and current queue status.	**Pass**: The patient's own token and appointment details were displayed. Other patients' tokens could not be viewed. Waiting and Cancelled statuses were observed.
Patients may need to change appointments or book a follow-up.	FR-2, FR-4	SC-05	SC-05 My Appointments	**Partial Pass**: Cancellation updated the status to Cancelled. Rescheduling to another available slot generated a new token. Follow-up booking was **Not tested** because no follow-up was needed during testing.
Admins need to maintain accurate doctor information and schedules.	FR-5	SC-11 Doctors & Schedules	Add, edit, and remove (deactivate) doctor records; manage clinic days and appointment times.	**Pass**: Admins could edit doctor information and schedules, and the updated information appeared in the patient view. Permanent removal of doctors with active appointments was prevented.
Admins need to monitor appointments and patient accounts without changing	FR-6, FR-7	SC-08 Admin Dashboard → SC-09 Admin Appointments → SC-10 Patient Accounts	View dashboard statistics and appointment records; Ban or Restore patient accounts.	**Pass**: Admins could monitor dashboard information and view appointment/token records without

appointment or queue statuses.				editing them. Admins could view account status, Ban/Restore accounts, and banned patients were blocked from access.
Patients need to create accounts and manage their personal information.	FR-8	SC-01 Sign In / Register → SC-07 My Profile	Register, log in, and update profile information.	**Pass**: Sign-in and account-creation options were available. Patients could edit their name, email and phone number. Banned accounts were restricted.
Patients and Admins need to receive and manage important system updates.	FR-9	SC-03 Patient Overview / SC-08 Admin Dashboard → Notification Panel	Display notifications, unread count, notification list, and Mark as Read actions.	**Partial Pass**: Users could view notifications and unread counts, and mark all as read. A notification after rescheduling was observed. Other notification types and the exclusion of visit-reason information still need verification.